

# What Rotation Equivariance Changes in Histopathology VAEs: Reconstruction, Label Efficiency, and Decoder Geometry

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**Abstract**—Histopathology has abundant unlabeled image patches, scarce expert labels, and no preferred tissue orientation. These properties motivate rotation-aware self-supervised representations, but architectural equivariance does not by itself guarantee better reconstructions, downstream performance, or a simple latent pose coordinate. We compare two denoising variational autoencoders (VAEs) trained on the same ovarian-cancer patches: a conventional convolutional VAE and a continuous-rotation-equivariant  $SO(2)$  VAE. Both use a spatial  $16 \times 32 \times 32$  Gaussian latent and are frozen after 60,000 updates. We evaluate reconstruction on 67,138 patches from 23 held-out whole-slide images (WSIs), five-class WSI diagnosis, three-class tissue recognition under five label budgets, and transformation consistency of the frozen latent-decoder maps. The normal VAE has a slightly lower mean absolute reconstruction error, but the paired WSI-bootstrap interval for the primary difference includes zero. The  $SO(2)$  representation yields higher point estimates for WSI macro-F1 (0.483 vs. 0.389) and accuracy (0.609 vs. 0.522), without a conclusive macro-F1 interval. In tissue recognition, it is better at four low-label budgets, but only 500 labels per class has a simultaneous interval excluding zero; the highest budget favors the normal VAE. The strongest mechanistic result is more specific: applying the prescribed spatial action to the  $SO(2)$  latent before decoding realizes exact quarter-turn image rotations at approximately  $10^{-6}$  RMS error, while raw encoder posterior means are not strictly equivariant. Thus, rotation-aware structure is clearly visible in the decoder, yet it does not imply universal task improvement or content-pose factorization. This endpoint-specific analysis provides a cautious template for evaluating equivariant generative representations in digital pathology.

**Index Terms**—histopathology, variational autoencoder, rotation equivariance,  $SO(2)$ , whole-slide image classification, label efficiency

## I. INTRODUCTION

Whole-slide histopathology combines gigapixel images with labels that require specialist review. Patch-based representation learning is therefore attractive: large quantities of unlabeled tissue can train an encoder before a smaller supervised model is fitted for diagnosis or tissue recognition [1]–[3]. A second property makes pathology a natural setting for symmetry-aware learning. Unlike upright natural objects, many local tissue structures retain their biological meaning after in-plane rotation.

Equivariant neural networks encode a transformation law rather than learning it only from augmentation [4], [5]. Steerable convolutions extend this principle to continuous planar rotations and reflections [6]; rotation-equivariant classifiers have also been studied directly in digital pathology [7]. In principle, a correct inductive bias can reduce the number of

examples needed to learn a target function [8]. In practice, however, three questions are often conflated:

- 1) Does an equivariant autoencoder reconstruct unseen tissue better?
- 2) Are its frozen representations more useful with limited labels?
- 3) Where is the prescribed symmetry visible in a trained encoder-decoder?

The questions are related but not equivalent. A decoder may commute with a group action even when the encoder selects different representatives. A useful downstream representation need not minimize pixel error. Conversely, a clean architectural transformation law need not produce a low-dimensional latent pose coordinate after training.

This work separates those claims in a matched experiment. We train a normal convolutional denoising VAE and a repo-owned continuous- $SO(2)$  steerable VAE on the same  $256 \times 256$  histopathology patches. The models share the data, spatial latent shape, denoising objective, optimizer policy, update budget, and evaluation splits. We then freeze both checkpoints and compare: (i) sealed reconstruction, (ii) WSI diagnosis with the same multiple-instance learner, (iii) tissue classification over nested label budgets, and (iv) exact and interpolated rotation consistency. We additionally examine the decoder pullback metric locally, because Euclidean latent differences can exaggerate directions that barely affect the decoded image [9].

Our contribution is not a claim that one model wins every endpoint. Instead, we provide a controlled and statistically explicit account of where continuous rotation structure helps, where it does not, and which equivariance claims are actually supported by the frozen checkpoints.

## II. RELATED WORK

### A. Weakly Supervised Histopathology

WSIs are commonly represented as bags of patch features because processing the complete slide at native resolution is infeasible. Attention-based multiple instance learning (MIL) learns a slide-level predictor from bag labels [10], and later computational-pathology systems have shown that such weak supervision can be data efficient [11]. Tellez *et al.* demonstrated a complementary route: learn an unsupervised compression model on image patches and use the resulting codes

for gigapixel prediction [3]. Our evaluation follows this two-stage logic, but holds the downstream learners fixed while changing the frozen representation.

### B. Equivariance and Generative Representations

For a group  $G$ , an encoder  $E$ , decoder  $D$ , input action  $T_g$ , and latent action  $\rho_g$ , strict equivariance and decoder covariance are different conditions:

$$E(T_g x) = \rho_g E(x), \quad (1)$$

$$D(\rho_g z) = T_g D(z). \quad (2)$$

The first constrains the representative produced by the encoder; the second constrains how a latent transforms when decoded. Group-equivariant and steerable CNNs provide constructions for these transformation laws [4]–[6]. VAEs provide a probabilistic spatial representation through posterior parameters  $\mu$  and  $\log \sigma^2$  and the reparameterization trick [12]. Our study evaluates both posterior and decoded transformation behavior instead of treating architectural equivariance as sufficient empirical evidence.

## III. METHODS

### A. Dataset and Split Discipline

We use ovarian-cancer histopathology from the UBC-OCEAN resource [13]. The unsupervised source contains 300,000 training and 30,000 validation patches from 322 and 39 non-TMA WSIs, respectively. Patches are RGB,  $256 \times 256$ , and normalized to  $[-1, 1]$ . The training corruption changes hematoxylin/eosin stain coordinates and adds Gaussian noise with probability 0.3. Twenty-five validation patches, five per source diagnosis, were fixed before model comparison for visual and rotation analyses.

The sealed evaluation comprises 23 WSIs disjoint from the 106 training and 23 development WSIs used by downstream classifiers. It contains 67,138 foreground patches for reconstruction. The WSI task predicts five ovarian diagnoses with supports 5/6/8/2/2. The tissue task uses 31,572 patches from the same 23 WSIs: 21,796 tumor, 8,500 stroma, and 1,276 necrosis patches. Necrosis occurs in only five WSIs. Sealed results were never used to select checkpoints, architectures, label budgets, or stopping rules.

### B. Matched Variational Autoencoders

Both VAEs map an input to spatial posterior parameters  $\mu, \log \sigma^2 \in \mathbb{R}^{16 \times 32 \times 32}$  and decode a sampled latent to RGB. They share 43 learned convolutions, the spatial schedule  $256 \rightarrow 128 \rightarrow 64 \rightarrow 32$ , fixed anti-aliased downsampling, bilinear upsampling followed by convolution, and a raw RGB output. The objective is

$$\mathcal{L} = \|\hat{x} - x\|_1 + 0.1(1 - \text{SSIM}(\hat{x}, x)) + \beta D_{\text{KL}}(q_\phi(z | \hat{x}) \| \mathcal{N}(0, I)), \quad (3)$$

with  $\beta = 0.01$  after warmup. AdamW, learning-rate scheduling, corruption, training order, and the 60,000-update budget are matched.

TABLE I  
FROZEN REPRESENTATION CONTRACT.

| Property            | Normal VAE               | SO(2) VAE                |
|---------------------|--------------------------|--------------------------|
| Updates             | 60,000                   | 60,000                   |
| Latent $\mu$ shape  | $16 \times 32 \times 32$ | $16 \times 32 \times 32$ |
| Learned parameters  | 3,958,435                | 1,180,035                |
| Training patches    | 300,000                  | 300,000                  |
| Rotation constraint | none                     | continuous SO(2)         |
| Downstream state    | frozen $\mu$             | frozen $\mu$             |

The normal VAE uses ordinary convolutions, GroupNorm, and scalar gates. The SO(2) VAE replaces these operations with continuous steerable convolutions, field-aware normalization, and radial gates over scalar and first-order field types. Its final posterior comprises 16 spatial scalar fields. The comparison is intentionally capacity-conservative: the SO(2) model has 1,180,035 learned parameters versus 3,958,435 for the normal model (29.8%). This avoids giving the equivariant model more capacity, but it also means symmetry and parameter count are not causally separated.

### C. Downstream Evaluation

For WSI diagnosis, every foreground patch is represented by deterministic posterior  $\mu$ . The shared 1.51M-parameter MIL classifier first compresses each latent map with a trainable convolutional patch encoder. Two local attention blocks exchange information within a radius-two patch graph, while a learned global stream summarizes the complete WSI for five-class prediction. The two representation branches receive identical WSI membership, coordinate graphs, initial classifier weights, epoch order, optimizer, and validation schedule. Each branch independently selects its highest development macro-F1 checkpoint.

For tissue recognition, a smaller shared convolutional classifier predicts tumor, stroma, or necrosis from frozen patch latents. We use nested, class-balanced training subsets of 250, 500, 1,000, 2,500, and 5,671 labels per class. Each budget begins from identical weights for both representations. Natural-distribution validation selects the checkpoint; the complete sealed test distribution is evaluated once.

### D. Reconstruction and Rotation Geometry

Reconstruction is evaluated with MAE and MSE in the normalized domain and with PSNR and SSIM after clamping to  $[0, 1]$ . The primary endpoint is mean MAE.

For rotations, let  $R_k$  be an exact array rotation by  $90k$  degrees and let  $\rho_k$  rotate every spatial latent channel in the same way. We distinguish the encoded cycle  $e_k = E_\mu(R_k x)$  from the prescribed cycle  $a_k = \rho_k E_\mu(x)$ . The decoder-action error is

$$\epsilon_D(k, x) = \text{RMS}[D(a_k) - R_k D(e_0)]. \quad (4)$$

We also use an interpolated  $0, 5, \dots, 355$  degree sweep. Its normalized action ratio divides the action error by the error produced when the latent is left untransformed; 1 is the no-action baseline. The canonicalization ratio similarly compares

TABLE II  
SEALED RECONSTRUCTION OVER 67,138 PATCHES. VALUES ARE MEAN (POPULATION SD).

| Metric    | Normal VAE      | SO(2) VAE       |
|-----------|-----------------|-----------------|
| MAE ↓     | 0.0629 (0.0239) | 0.0642 (0.0223) |
| MSE ↓     | 0.0080 (0.0074) | 0.0083 (0.0067) |
| PSNR dB ↑ | 27.923 (2.721)  | 27.658 (2.595)  |
| SSIM ↑    | 0.7575 (0.0842) | 0.7436 (0.0862) |

$D(\rho_{-\theta}E_{\mu}(R_{\theta}x))$  with  $D(E_{\mu}(x))$  relative to leaving the rotated embedding unchanged. A preregistered absolute target of 0.5 asks whether the action removes at least half of the baseline error.

Finally, at fixed validation ranks 0 and 12 we estimate the local decoder pullback  $G(z) = J_D(z)^{\top} J_D(z)$  without materializing the Jacobian. Direct JVP/VJP products and stochastic Lanczos quadrature with 64 probes and depth 64 estimate the trace and dimensions required for 95% and 99% spectral mass. This is a two-anchor diagnostic, not a population estimate or proof of a global manifold.

#### E. Uncertainty and Claim Boundaries

Reconstruction differences use 10,000 paired WSI-cluster bootstrap replicates; patch standard deviations describe dispersion only. WSI diagnosis uses a diagnosis-stratified paired bootstrap over 23 WSIs. Tissue differences use a WSI bootstrap stratified by observed tissue support and simultaneous 95% intervals across the five label budgets. These intervals do not cover new VAE or classifier training seeds. Rotation analyses use the fixed validation set; their WSI-resampled intervals are descriptive and do not support population inference.

### IV. RESULTS

#### A. Reconstruction Fidelity

Both models preserve the major tissue structures and staining patterns on the same predetermined validation patches (Fig. 1). On the sealed set, the normal VAE has slightly better point estimates for all four reconstruction metrics (Table II). Primary MAE is 0.062872 for normal and 0.064230 for SO(2). The paired difference normal-minus-SO(2) is  $-0.001358$ , with 95% CI  $[-0.002213, 0.000046]$ . Because the interval includes zero, the primary reconstruction difference is not established. PSNR and SSIM favor the normal model descriptively, but remain secondary endpoints.

#### B. WSI Diagnosis and Tissue Label Efficiency

On 23 sealed WSIs, SO(2) has higher point estimates for macro-F1 and accuracy, while balanced accuracy is essentially unchanged (Fig. 2). Macro-F1 is 0.483 versus 0.389, but the 95% interval for normal-minus-SO(2),  $[-0.207, 0.050]$ , includes zero. Accuracy is 0.609 versus 0.522 with interval  $[-0.217, 0.043]$ . Mean cross-entropy is lower for SO(2) (0.993 versus 1.323), and its paired interval favors SO(2); this is a secondary endpoint. Class-level conclusions are fragile: LGSC

TABLE III  
SEALED DOWNSTREAM MACRO-F1. TISSUE INTERVALS ARE SIMULTANEOUS ACROSS BUDGETS AND ARE REPORTED FOR NORMAL-MINUS-SO(2).

| Task/budget   | Normal | SO(2) | Difference CI      |
|---------------|--------|-------|--------------------|
| WSI diagnosis | 0.389  | 0.483 | $[-0.207, 0.050]$  |
| Tissue 250    | 0.508  | 0.523 | $[-0.072, 0.041]$  |
| Tissue 500    | 0.511  | 0.575 | $[-0.121, -0.008]$ |
| Tissue 1,000  | 0.582  | 0.598 | $[-0.072, 0.041]$  |
| Tissue 2,500  | 0.701  | 0.714 | $[-0.069, 0.044]$  |
| Tissue 5,671  | 0.766  | 0.710 | $[-0.001, 0.112]$  |

and MC contain only two WSIs each, and the two models fail on different rare classes.

The tissue task reveals a non-monotone interaction with label budget. SO(2) has higher macro-F1 at 250, 500, 1,000, and 2,500 labels per class. Only the 500-label difference has a simultaneous interval excluding zero: normal-minus-SO(2) =  $-0.064$ , 95% simultaneous CI  $[-0.121, -0.008]$ . At 5,671 labels per class, the normal representation is higher (0.766 versus 0.710), although the simultaneous interval nearly touches zero. The normalized area under the log-budget learning curve is 0.615 for normal and 0.631 for SO(2); its simultaneous difference interval includes zero. Therefore, the results support one budget-specific low-label advantage, not universal sample-efficiency dominance.

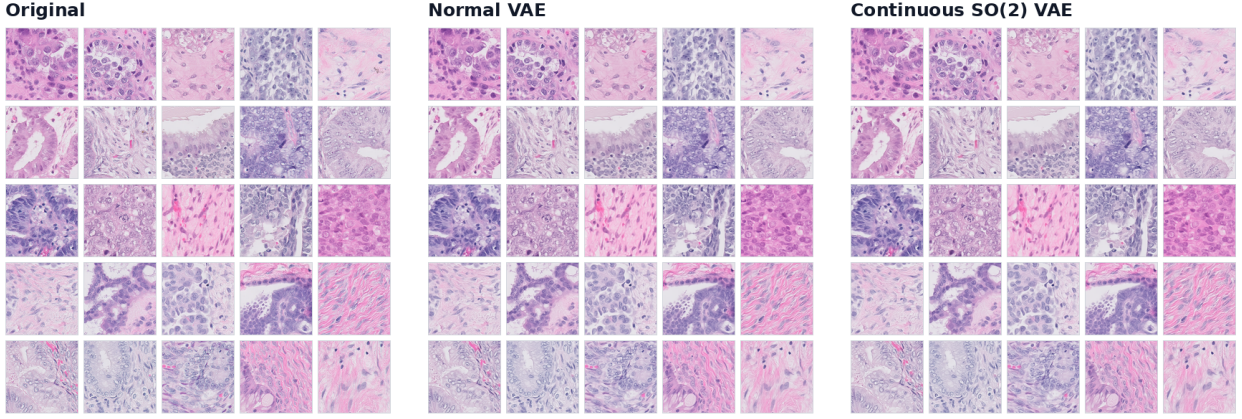
#### C. Decoder Equivariance Is Stronger Than Encoder Equivariance

The dense interpolated sweep favors SO(2) consistently in route agreement. Median decoded-action ratios are 0.550 for SO(2) and 0.713 for normal; canonicalization ratios are 0.513 and 0.661. The SO(2) value is lower on all 25 paired patches for both metrics, and the WSI-resampled descriptive intervals for the paired median differences exclude zero. However, the SO(2) medians remain just above the absolute 0.5 target, so this dense interpolation experiment does not by itself verify exact continuous equivariance.

The exact quarter-turn control is qualitatively different. At 90, 180, and 270 degrees, the prescribed latent action of the SO(2) checkpoint commutes with decoding at approximately  $10^{-6}$  image RMS, compared with normal-VAE medians of 0.165–0.190. The corresponding normalized action ratio aggregates to  $2.18 \times 10^{-6}$  for SO(2) and 0.709 for normal. This verifies a shared, parameter-free  $C_4$  decoder action on all 25 fixed patches (Fig. 3). It does not verify all continuous angles, encoder equivariance, reflections, or complete  $D_4/O(2)$  symmetry.

Indeed, raw posterior behavior is less favorable. For exact quarter-turns, the relative discrepancies between  $E_{\mu}(R_k x)$  and  $\rho_k E_{\mu}(x)$  are not lower for SO(2). Thus, Eqs. (1) and (2) must not be merged: the decoder realizes the prescribed cardinal action while the encoder posterior mean can select a different latent representative.

## Final reconstructions — same predetermined fixed validation 25



Ordered by frozen selector rank (five patches per source histotype); FP16 archives shown after  $[-1,1] \rightarrow [0,1]$  clamping.

Fig. 1. Originals and final reconstructions for the same 25 validation patches fixed before analysis. Visual inspection is descriptive; the sealed quantitative evaluation uses 67,138 patches from 23 WSIs.

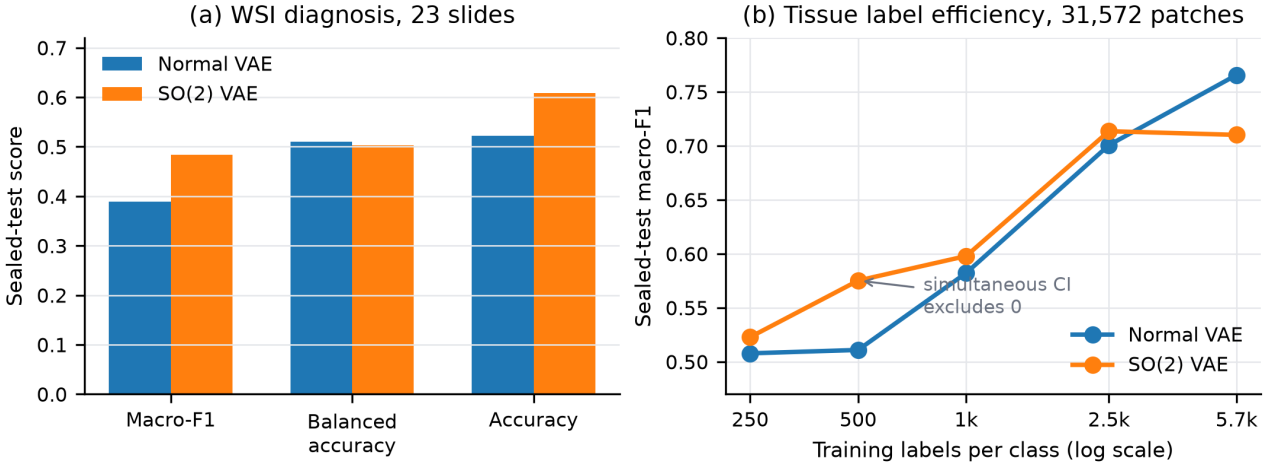


Fig. 2. Frozen-representation downstream results. (a) WSI diagnosis point estimates on 23 sealed slides. (b) Tissue macro-F1 on 31,572 sealed patches; training subsets are nested and class balanced. Only 500 labels per class has a simultaneous difference interval excluding zero.

### D. Local Decoder-Visible Geometry

The pullback analysis supports a narrower claim about decoder-visible complexity. Across ranks 0 and 12 and four exact angles, the estimated  $SO(2)$ /normal trace ratio ranges from 0.895 to 0.959. The  $SO(2)$  estimate also requires fewer dimensions to reach both 95% and 99% spectral mass in all eight matched states. At rank 0, normal  $d_{95}$  ranges 4,886–4,928 while  $SO(2)$  ranges 4,565–4,581; at rank 12, the ranges are 3,835–4,036 and 3,726–3,750. These observations are consistent with a somewhat more compact local decoder-visible spectrum, but they come from only two anchors. They do not establish a population effect, a continuous rotation orbit, a global quotient, or low-dimensional disentanglement.

### V. DISCUSSION

The results separate three effects that would be obscured by a single summary score. First, the normal VAE retains a small descriptive reconstruction advantage. The primary interval crosses zero, but the consistent secondary metrics warn against assuming that stronger architectural structure must improve pixel fidelity.

Second, the  $SO(2)$  representation shows promising but uneven downstream behavior. Its WSI point estimates are better for macro-F1 and accuracy, and it has a clear simultaneous advantage at one low-label tissue budget. Yet these signals do not persist at the highest tissue budget, and the global learning curve comparison is inconclusive. This pattern is compatible with an inductive bias that can help in a restricted low-data regime, but one training seed and 23 test WSIs cannot isolate

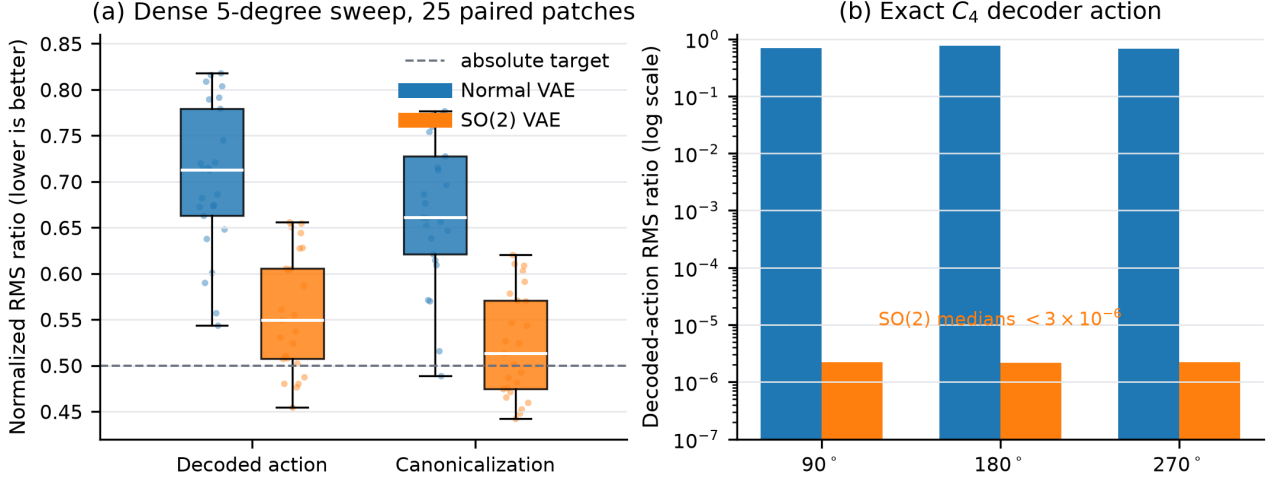


Fig. 3. Transformation consistency on the fixed validation set. (a) Interpolated 5-degree sweep; lower normalized RMS ratios indicate better route agreement, and 0.5 is the preregistered absolute target. (b) Exact cardinal rotations on a logarithmic scale. The SO(2) decoder realizes the prescribed  $C_4$  action almost exactly.

TABLE IV  
EVIDENCE MAP. EACH CONCLUSION IS INTENTIONALLY LIMITED TO THE MEASUREMENT THAT SUPPORTS IT.

| Question                         | Main evidence                                   | Supported conclusion                                                             |
|----------------------------------|-------------------------------------------------|----------------------------------------------------------------------------------|
| Reconstruction fidelity          | 67,138 sealed patches; WSI-cluster bootstrap    | Normal is descriptively better; the primary MAE difference is not established.   |
| WSI representation utility       | 23 sealed WSIs under one shared MIL contract    | SO(2) has favorable point estimates, without a conclusive macro-F1 interval.     |
| Low-label tissue utility         | Five nested budgets with simultaneous intervals | One supported advantage at 500 labels/class; no global sample-efficiency claim.  |
| Decoder rotation action          | Fixed-25 exact quarter-turn control             | The SO(2) decoder realizes the prescribed $C_4$ action near numerical precision. |
| Encoder equivariance             | $E_\mu(R_k x)$ versus $\rho_k E_\mu(x)$         | Strict equivariance of the final posterior mean is not supported.                |
| Continuous/content-pose geometry | Dense interpolation and shared-action probes    | No verified continuous action, global quotient, or content-pose factorization.   |

that mechanism.

Third, the most decisive evidence is mechanistic rather than task-level. The SO(2) decoder realizes the prescribed exact  $C_4$  spatial action nearly perfectly. At the same time, the raw posterior mean does not satisfy the same strict encoder relation, dense interpolated ratios miss the absolute target, and previous low-dimensional shared-action probes fail to outperform identity. Consequently, the correct interpretation is not that the learned latent equals “content times angle.” A more conservative interpretation is that the decoder contains a coherent cardinal rotation mechanism while the encoder may choose different representatives in a high-dimensional, decoder-redundant space.

The parameter count deserves equal emphasis. The SO(2) VAE uses only 29.8% of the parameters of the normal VAE. This is an efficiency result, but also a causal confound: any difference may reflect symmetry, capacity, optimization, or their interaction. Our study therefore supports endpoint-specific comparison between these frozen checkpoints, not attribution to symmetry alone.

## VI. LIMITATIONS

The study has one training trajectory per VAE and one downstream trajectory per representation and budget. Bootstrap intervals measure sampling variation across observed WSIs, not initialization or training-seed uncertainty. The WSI test contains 23 slides, with only two examples in each of two diagnoses. Tissue evaluation contains many patches but only 23 biological clusters, and necrosis appears in five.

The reconstruction aggregation and bootstrap were finalized after later test labels had been opened, although neither VAE checkpoint was changed. Rotation geometry is post-hoc validation-only analysis on 25 fixed patches. Interpolated angles depend on bilinear resampling and do not inherit the numerical exactness of array quarter-turns. Pullback spectra cover two anchors only. Finally, the models differ substantially in parameter count, so the experiment does not identify the causal effect of equivariance independently of capacity.

## VII. CONCLUSION

A continuous-SO(2) VAE can expose meaningful rotation structure without winning every conventional endpoint. Here, the normal VAE reconstructs slightly better descriptively; the SO(2) representation has favorable but uncertain WSI results and one simultaneous low-label tissue advantage; and its decoder realizes exact cardinal rotations with near-numerical precision. Raw posterior means do not support strict encoder equivariance or a simple content–pose factorization. Evaluations should therefore report reconstruction, downstream utility, and transformation mechanism separately.

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